# Laxmi Poudel

U.S. Citizen | Eligible for U.S. Security Clearance

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## PROFESSIONAL SUMMARY

Software Engineer (B.S. Software Engineering) who builds full-stack features end to end (REST APIs, object-oriented design) and owns the test automation behind them. Shipped customer-facing software for an FAA air-traffic surveillance system on a Java/Spring, Angular, and PostgreSQL stack (2x faster queries, ~20% better data accuracy). Now builds the automated test software for a high-volume radar production program in LabVIEW, TestStand, and Python, tracing failures down to the fix (took a key step's first-time pass rate from 0% to 100%). Holds personal projects to the same standard: a Next.js/TypeScript PWA live on GitHub Pages backed by 570+ automated tests. Open to Software Engineer and SDET / test-automation roles.

## TECHNICAL SKILLS

**Languages:** Java, Python, TypeScript, JavaScript, C/C++, SQL, HTML/CSS

**Testing & Quality:** JUnit, Selenium, Playwright, Vitest, pytest, unit / integration / end-to-end testing, accessibility testing (axe-core), LabVIEW, TestStand, root-cause debugging

**Frameworks & Libraries:** Spring, Spring Boot, Angular, React, Next.js, Express, Flask, Tailwind, Maven, JSP, NumPy, SciPy, Log4j

**Databases:** PostgreSQL, MySQL, MongoDB, Firestore

**Cloud, DevOps & Tools:** Docker, Kubernetes, Azure, Firebase, GitHub Actions, CI/CD, Jenkins, Git, Jira, Wildfly, IntelliJ, VS Code, Eclipse

**Platforms & Methods:** Linux, Windows Server, Agile / Scrum, REST APIs, FPGA programming, PMIC programming, T&M hardware (signal generators/analyzers, power meters, multimeters)

## PROFESSIONAL EXPERIENCE

### SRCTec

August 2025 - Present

**Test Engineer** | North Syracuse, NY

- Build and maintain the automated test software for a radar production program in LabVIEW, TestStand, and Python, developing in-house test applications and integrating third-party packages to support the program's test stations.
- Root-caused a system-level FPGA programming failure and added a post-programming power cycle, lifting that step's first-time pass rate from 0% to 100% and cutting it from two test runs to one (~50% less test time on the step).
- Restored production after a Windows 10-to-11 upgrade broke a board-level test's detection of TI Fusion for PMIC programming, tracing the regression to the OS change and shipping a hotfix.
- Led migration of all program test software from LabVIEW 2016 to 2021 across the test stations and 8 software packages in under 3 months, fixing the version-jump breakages it surfaced.

### SRCTec

June 2024 - August 2024

**Test Engineering Intern** | North Syracuse, NY

- Cut manual effort ~40% by automating radar test delay-line and RF-switch settings with Python over HTTPS.
- Resolved 15 issues across multiple libraries and upgraded Python 2.7 to 3.10, clearing backlog and improving testing speed ~25%.
- Migrated radar calibration software from Linux to Windows Server, improving performance and compatibility.

### Saab Inc.

January 2023 - August 2023

**Software Engineering Co-op** | East Syracuse, NY

- Launched an Air Situation Display (ASD) feature letting users view all aircraft within a radius on click, replacing a top-aircraft-only view.
- Worked with the FAA to track objects via an Angular front end and Java back end, integrating the SysMon module into the Cooperative Surveillance System (CSS) through front-end/back-end API integration.
- Improved data accuracy ~20% by filtering low-quality ADS-B and MLAT data, and optimized PostgreSQL queries for ~2x faster retrieval; delivered in an Agile/Scrum team.

## PROJECTS

### Nepal x Japan Trip Planner

June 2026 - Present

**Solo build, shipped and live** | powan55.github.io/trip_planner | github.com/Powan55/trip_planner

- Built and shipped an offline-first trip planning PWA (Next.js 16, React 19, TypeScript, Tailwind, MapLibre) as a static export on GitHub Pages; a hand-rolled service worker precaches the app shell, so it installs and runs with no connection.
- Kept data on the device (localStorage, IndexedDB) with an optional Firestore sync layer, and gated releases in GitHub Actions behind 107 Playwright spec files (570+ end-to-end cases), 311 Vitest files, and axe-core accessibility checks.

### Single Pane of Glass (SPOG)

August 2024 - May 2025

**Senior Capstone sponsored by LenelS2 (Honeywell)** | RIT

- Extended the sponsor's internal dashboard unifying cost, performance, and alerting across AWS, Azure, and GCP with a 5-person team over a full academic year; migrated the React front end from JavaScript to TypeScript and standardized on Material UI.
- Implemented login/authentication, user creation, and role-based user privileges (the "Persona" access model) in the TypeScript/Express People Service (REST APIs, MongoDB); rendered cloud metrics and logs through Grafana components.

## EDUCATION

**B.S. in Software Engineering** | Rochester Institute of Technology (RIT)

Aug 2021 - May 2025

Rochester, NY